

Linear Algebra — Week 02

Vector Spaces of n -tuples, Matrices & Functions; Matrix Operations

Why this matters

Last week we studied fields and the abstract axioms of a vector space. This week we meet the **three vector spaces used constantly throughout the course**: n -tuples, matrices, and functions. For each one, we need to pin down exactly what its elements are, how to add them, how to scalar-multiply them, and — the question students most often underestimate — **when two of them count as equal**. Matrices, in particular, are more than notation for keeping numbers organized: they are objects you can do arithmetic with (Siefken, Appendix 3).

DEFINITION The Vector Space F^n

The vector space of F n -tuples is

$$F^n = \{(x_1, x_2, \dots, x_n) : x_i \in F\}.$$

Addition and scalar multiplication are defined coordinatewise:

$$x + y = (x_1, \dots, x_n) + (y_1, \dots, y_n) := (x_1 + y_1, \dots, x_n + y_n), \quad cx := (cx_1, \dots, cx_n).$$

Let $x, y \in F^n$. Then $x = y$ if and only if $x_i = y_i$ for all i . (Kielstra, 0A: n -tuples)

Key computational fact: $x + z$ is only defined when x and z live in the *same* F^n — i.e. have the same number of coordinates. Addition between F^n and F^m with $n \neq m$ is simply not defined; there is no “padding” or “truncating.” (Kielstra, 0A: n -tuples)

DEFINITION The Vector Space of $n \times k$ Matrices $M_{n \times k}(F)$

The vector space of $n \times k$ matrices is the set of all rectangular arrays with entries $A_{ij} \in F$, $1 \leq i \leq n$, $1 \leq j \leq k$. For $A, B \in M_{n \times k}(F)$:

$$(A + B)_{ij} := A_{ij} + B_{ij}, \quad (cA)_{ij} := cA_{ij}.$$

$A = B$ if and only if (1) A and B have the same dimensions, **and** (2) $A_{ij} = B_{ij}$ for all i, j . (Kielstra, 0B: $n \times k$ Matrices)

Matrix addition only makes sense for matrices of the same shape; any matrix can be scalar-multiplied by any scalar, regardless of shape. (Siefken, Appendix 3)

DEFINITION Matrix Notation, Shape, and Special Matrices

A matrix is described by its **shape** (rows $\times$ columns) and its **entries**. An $m \times n$ matrix $A = [a_{ij}]$ has a_{ij} in row i , column j . The **diagonal** of A consists of entries a_{ij} with $i = j$; the **upper triangle** consists of entries with $j \geq i$; the **lower triangle** consists of entries with $j \leq i$. (Siefken, Appendix 3)

- A matrix is **triangular** (upper or lower) if all its non-zero entries lie in the corresponding triangle.
- A matrix is **square** if it has the same number of rows as columns.
- A square matrix is **diagonal** if its only non-zero entries lie on the diagonal.
- A square matrix $A = [a_{ij}]$ is **symmetric** if $a_{ij} = a_{ji}$ for all i, j (and **skew-symmetric** if $a_{ij} = -a_{ji}$).

- A **zero matrix** has every entry equal to 0.
- An **identity matrix** $I_{n \times n}$ (or just I) is a square matrix with 1's on the diagonal and 0's everywhere else.

(Siefken, Appendix 3)

DEFINITION Transpose of a Matrix

Let $A \in M_{n \times k}(F)$ with entries A_{ij} . The **transpose** of A , denoted A^T , is the $k \times n$ matrix defined by

$$(A^T)_{ij} := A_{ji}.$$

That is, the transpose is obtained by interchanging the rows and columns of A . (Kielstra, 0B: $n \times k$ Matrices)

Key computational fact: because transposing changes the shape of a matrix (an $n \times k$ matrix becomes $k \times n$), you must re-check dimension compatibility *after* transposing before adding two matrices — the original shapes being compatible (or not) tells you nothing about the transposed shapes.

DEFINITION Matrix-Vector Multiplication

Let A be an $m \times n$ matrix with columns $\vec{c}_1, \dots, \vec{c}_n$ and rows $\vec{r}_1, \dots, \vec{r}_m$ (written as column vectors), and let $\vec{x} \in F^n$.

- **Column picture:** $A\vec{x}$ is the linear combination of the columns of A with coefficients $x_1, \dots, x_n$:

$$A\vec{x} = x_1\vec{c}_1 + x_2\vec{c}_2 + \dots + x_n\vec{c}_n.$$

- **Row picture:** $A\vec{x}$ is the vector whose i th coordinate is the dot product of row i of A with $\vec{x}$:

$$A\vec{x} = \begin{bmatrix} \vec{r}_1 \cdot \vec{x} \\ \vec{r}_2 \cdot \vec{x} \\ \vdots \\ \vec{r}_m \cdot \vec{x} \end{bmatrix}.$$

The two pictures always give the same answer. $A\vec{x}$ is only defined if $\vec{x}$ has n coordinates (matching the number of *columns* of A); the result has m coordinates (the number of *rows* of A). (Siefken, Appendix 3)

DEFINITION Matrix-Matrix Multiplication

Let C be an $m \times n$ matrix and A an $n \times k$ matrix with columns $\vec{a}_1, \dots, \vec{a}_k$. Then CA is the $m \times k$ matrix whose columns are $C\vec{a}_1, \dots, C\vec{a}_k$ — i.e. matrix-matrix multiplication is built out of repeated matrix-vector multiplication. Equivalently, if C has rows $\vec{r}_1, \dots, \vec{r}_m$, the (i, j) entry of CA is $\vec{r}_i \cdot \vec{a}_j$ (row i of C dotted with column j of A).

For CA to be defined, the number of columns of C must equal the number of rows of A ; the resulting matrix has (rows of C) $\times$ (columns of A). (Siefken, Appendix 3)

KEY FACT Properties of Matrix Algebra

For $A, B, C \in M_{n \times n}(F)$ and $\alpha \in F$:

$$(AB)C = A(BC), \quad A(B + C) = AB + AC, \quad (A + B)C = AC + BC, \\ IA = AI = A, \quad 0A = A0 = 0.$$

Matrix multiplication is, in general, not commutative: $AB \neq BA$ in general, even for square matrices of the same size. We write matrix-vector and matrix-matrix products by adjacency ($A\vec{v}$, AB) — never with $\cdot$ or $\times$, which are reserved for the dot product and cross product. (Siefken, Appendix 3)

DEFINITION The Vector Space of F -Valued Functions $\mathcal{F}(F)$

$$\mathcal{F}(F) = \{f : F \rightarrow F\}.$$

For $f, g \in \mathcal{F}(F)$ and $c \in F$:

$$(f + g)(x) := f(x) + g(x), \quad (cf)(x) := cf(x).$$

Two functions f, g are equal if and only if $f(x) = g(x)$ **for all** $x \in F$ — function equality is checked *pointwise*, not by “looking similar” or agreeing on part of the domain. (Kielstra, 0C: *F*-Valued Functions)

Definition: Composition. $(f \circ g)(x) := f(g(x))$. (Kielstra, 0C: *F*-Valued Functions)

KEY FACT Note: Generic Elements Simplify Proofs

Proving an identity about F^n , matrices, or functions by expanding every coordinate/entry/input leads to long, repetitive expressions. Instead, fix an arbitrary index i (for F^n), an arbitrary pair (i, j) (for matrices), or an arbitrary input x (for functions), and prove the identity holds for that single generic piece. Since the choice was arbitrary, the identity then holds everywhere at once. This is the standard proof strategy used throughout this unit. (Kielstra, 0A/0B/0C)

Worked Examples

EXAMPLE 1

Equality of n -tuples

Let $x = (2, -1, 5)$, $y = (2, -1, 5)$. Verify $x = y$ using the definition of equality in F^3 . By definition, $x = y$ if and only if $x_i = y_i$ for $i = 1, 2, 3$. Here $x_1 = 2 = y_1$, $x_2 = -1 = y_2$, $x_3 = 5 = y_3$. Since $x_i = y_i$ for all i , we conclude $x = y$. ■

EXAMPLE 2

Can these n -tuple operations be computed?

Let $x = (1, 2, 3)$, $y = (4, 5, 6)$, $z = (1, 2)$. Determine whether $x + y$ and $x + z$ can be computed.

x and y are both in F^3 , so their sum is defined:

$$x + y = (1 + 4, 2 + 5, 3 + 6) = (5, 7, 9).$$

x is in F^3 while z is in F^2 . Since vector addition on F^n is only defined between two elements of the *same* F^n , and $3 \neq 2$, the sum $x + z$ is **not defined**. ■

EXAMPLE 3

Distributivity in F^n , generic-coordinate proof

Let $x, y \in F^n$ and $c \in F$. Prove $c(x + y) = cx + cy$.

Rather than expand all n coordinates at once, fix an arbitrary index i with $1 \leq i \leq n$ and compare the i th entries of both sides:

$$\begin{aligned}
(c(x + y))_i &= c(x + y)_i \quad (\text{def. of scalar mult. in } F^n) \\
&= c(x_i + y_i) \quad (\text{def. of vector addition in } F^n) \\
&= cx_i + cy_i \quad (\text{distributivity in the field } F) \\
&= (cx)_i + (cy)_i \quad (\text{def. of scalar mult. in } F^n) \\
&= (cx + cy)_i \quad (\text{def. of vector addition in } F^n).
\end{aligned}$$

Since $(c(x + y))_i = (cx + cy)_i$ for every $i = 1, \dots, n$, the two n -tuples agree in every coordinate, so $c(x + y) = cx + cy$. ■

EXAMPLE 4

Matrix addition and scalar multiplication

Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$, $C = \begin{bmatrix} -1 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix}$. Compute $2A + B$ and $A + 3C$, if possible.

First check shapes. A is 2×3 , so it can only be added to another 2×3 matrix. B is 2×2 , so $2A + B$ is **not defined** — the shapes don't match, no matter what scalar multiple of A we take. C , however, is 2×3 , so $A + 3C$ is defined. We first scalar-multiply C entrywise, then add entrywise:

$$3C = \begin{bmatrix} -3 & 0 & -3 \\ 6 & 3 & 6 \end{bmatrix}, \quad A + 3C = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} + \begin{bmatrix} -3 & 0 & -3 \\ 6 & 3 & 6 \end{bmatrix} = \begin{bmatrix} 1-3 & 2+0 & 3-3 \\ 4+6 & 5+3 & 6+6 \end{bmatrix} = \begin{bmatrix} -2 & 2 & 0 \\ 10 & 8 & 12 \end{bmatrix}. \quad \blacksquare$$

(Siefken, Appendix 3)

EXAMPLE 5

Matrix transpose and a combined expression

Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ (2×3) and $B = \begin{bmatrix} 1 & 0 \\ -1 & 2 \\ 3 & 1 \end{bmatrix}$ (3×2). Compute A^T , and determine whether

$A^T + B$ is defined; if so, compute it.

Transposing A swaps its rows and columns, turning the 2×3 matrix into a 3×2 matrix:

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

Now A^T is 3×2 and B is 3×2 — the shapes match, so $A^T + B$ **is** defined:

$$A^T + B = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ -1 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 1+1 & 4+0 \\ 2-1 & 5+2 \\ 3+3 & 6+1 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 1 & 7 \\ 6 & 7 \end{bmatrix}. \quad \blacksquare$$

(Kielstra, 0B: $n \times k$ Matrices)

EXAMPLE 6

Matrix-vector multiplication: column and row pictures

Let $B = \begin{bmatrix} 1 & 2 \\ -2 & 3 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$. Compute $B\vec{v}$ using the column picture, then verify with the row picture.

Column picture. The columns of B are $\vec{c}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ and $\vec{c}_2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$. We take the linear combination with coefficients from $\vec{v}$:

$$B\vec{v} = 4 \begin{bmatrix} 1 \\ -2 \end{bmatrix} + 3 \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ -8 \end{bmatrix} + \begin{bmatrix} 6 \\ 9 \end{bmatrix} = \begin{bmatrix} 10 \\ 1 \end{bmatrix}.$$

Row picture. The rows of B (written as column vectors) are $\vec{r}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{r}_2 = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$. Each output coordinate is a dot product with $\vec{v}$:

$$B\vec{v} = \begin{bmatrix} \vec{r}_1 \cdot \vec{v} \\ \vec{r}_2 \cdot \vec{v} \end{bmatrix} = \begin{bmatrix} (1)(4) + (2)(3) \\ (-2)(4) + (3)(3) \end{bmatrix} = \begin{bmatrix} 10 \\ 1 \end{bmatrix}.$$

Both pictures agree: $B\vec{v} = \begin{bmatrix} 10 \\ 1 \end{bmatrix}$. ■ (Siefken, Appendix 3)

EXAMPLE 7

Matrix-matrix multiplication and non-commutativity

Let $X = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 0 \end{bmatrix}$ and $Y = \begin{bmatrix} 2 & 3 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$. Compute XY and YX .

First check compatibility: X is 2×3 and Y is 3×2 . For XY : columns of X (3) match rows of Y (3), so XY is defined and will be 2×2 . Using the row-dot-column rule, the $(1, 1)$ entry of XY is row 1 of X dotted with column 1 of Y :

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = 2 + 2 + 3 = 7.$$

Computing every entry the same way gives

$$XY = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 7 & 5 \\ -1 & -1 \end{bmatrix}.$$

Now for YX : columns of Y (2) match rows of X (2), so YX is defined and will be 3×3 .

Computing entry by entry (e.g. the $(1, 1)$ entry is $\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 2$):

$$YX = \begin{bmatrix} 2 & 3 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 6 \\ 1 & 1 & 3 \\ 1 & 2 & 3 \end{bmatrix}.$$

Notice XY is 2×2 while YX is 3×3 — they aren't even the same shape, so certainly $XY \neq YX$. This illustrates that matrix multiplication is **not commutative**, and in fact AB and BA need not even have the same shape. ■ (Siefken, Appendix 3)

EXAMPLE 8

F-valued functions form a vector space; composition distributes over addition

Let $f, g, h \in \mathcal{F}(F)$. Prove $(f + g) \circ h = f \circ h + g \circ h$.

Fix an arbitrary $x \in F$ and compare the values of both sides at x :

$$\begin{aligned} ((f + g) \circ h)(x) &= (f + g)(h(x)) \quad (\text{def. of composition}) \\ &= f(h(x)) + g(h(x)) \quad (\text{def. of function addition, applied at the input } h(x)) \\ &= (f \circ h)(x) + (g \circ h)(x) \quad (\text{def. of composition, applied to each term}) \\ &= (f \circ h + g \circ h)(x) \quad (\text{def. of function addition}). \end{aligned}$$

Since this holds for every $x \in F$, and function equality is exactly agreement at every input, we conclude $(f + g) \circ h = f \circ h + g \circ h$. ■ (Kielstra, OC: F-Valued Functions)

EXAMPLE 9

Level: Foundational — matrix-vector multiplication, straightforward

Let $A = \begin{bmatrix} 3 & -1 \\ 0 & 2 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$. Compute $A\vec{v}$ using the column picture.

Step 1. Read off the columns of A : $\vec{c}_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$, $\vec{c}_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$.

Step 2. Form the linear combination with coefficients from $\vec{v} = (2, 5)$: $A\vec{v} = 2\vec{c}_1 + 5\vec{c}_2$.

Step 3. Scale each column: $2\vec{c}_1 = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$, $5\vec{c}_2 = \begin{bmatrix} -5 \\ 10 \end{bmatrix}$.

Step 4. Add: $A\vec{v} = \begin{bmatrix} 6 \\ 0 \end{bmatrix} + \begin{bmatrix} -5 \\ 10 \end{bmatrix} = \begin{bmatrix} 1 \\ 10 \end{bmatrix}$. ■

EXAMPLE 10

Level: Intermediate — a symmetric-matrix proof by generic entries

Let $A \in M_{n \times n}(F)$. Prove that $A + A^T$ is always symmetric.

Step 1. Let $S = A + A^T$. By definition, S is symmetric if $S_{ij} = S_{ji}$ for every i, j — so fix an arbitrary pair (i, j) and compare both sides.

Step 2. Expand the (i, j) entry using the definitions of matrix addition and transpose: $S_{ij} = (A + A^T)_{ij} = A_{ij} + (A^T)_{ij} = A_{ij} + A_{ji}$.

Step 3. Expand the (j, i) entry the same way: $S_{ji} = (A + A^T)_{ji} = A_{ji} + (A^T)_{ji} = A_{ji} + A_{ij}$.

Step 4. Compare: $S_{ij} = A_{ij} + A_{ji}$ and $S_{ji} = A_{ji} + A_{ij}$ are the same sum in the field F (addition is commutative), so $S_{ij} = S_{ji}$ for every i, j . Hence $A + A^T$ is symmetric. ■

EXAMPLE 11

Level: Challenge — matrix multiplication is not commutative, but IS associative here; a three-matrix computation

Let $X = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $Y = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $Z = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Compute $(XY)Z$ and $X(YZ)$, and confirm they agree (illustrating associativity, Fact on Page 1) even though $XY \neq YX$ in general.

Step 1. Compute XY : row i of X dotted with column j of Y . $X = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $Y = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$:
 $(XY)_{11} = (1)(0) + (0)(1) = 0$; $(XY)_{12} = (1)(1) + (0)(0) = 1$; $(XY)_{21} = (2)(0) + (1)(1) = 1$;
 $(XY)_{22} = (2)(1) + (1)(0) = 2$. So $XY = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}$.

Step 2. Compute $(XY)Z = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$: (1,1): $(0)(1) + (1)(0) = 0$; (1,2): $(0)(1) + (1)(1) = 1$; (2,1): $(1)(1) + (2)(0) = 1$; (2,2): $(1)(1) + (2)(1) = 3$. So $(XY)Z = \begin{bmatrix} 0 & 1 \\ 1 & 3 \end{bmatrix}$.

Step 3. Now compute $YZ = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$: (1,1): $(0)(1) + (1)(0) = 0$; (1,2): $(0)(1) + (1)(1) = 1$; (2,1): $(1)(1) + (0)(0) = 1$; (2,2): $(1)(1) + (0)(1) = 1$. So $YZ = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$.

Step 4. Compute $X(YZ) = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$: (1,1): $(1)(0) + (0)(1) = 0$; (1,2): $(1)(1) + (0)(1) = 1$; (2,1): $(2)(0) + (1)(1) = 1$; (2,2): $(2)(1) + (1)(1) = 3$. So $X(YZ) = \begin{bmatrix} 0 & 1 \\ 1 & 3 \end{bmatrix}$.

Step 5. Compare: $(XY)Z = \begin{bmatrix} 0 & 1 \\ 1 & 3 \end{bmatrix} = X(YZ)$ — they agree, confirming associativity for this triple, even though matrix multiplication is not commutative in general. ■

Practice Problems

Read the hint and the caution before you start each problem — not after.

***n*-tuples**

1. Let $x = (1, -1, 2)$ and $y = (0, 3, 4)$. Compute $x + y$ and $2x$.

HINT Apply coordinatewise addition and scalar multiplication one entry at a time.

CAUTION — before you solve Watch out: scale every coordinate by 2, and match corresponding coordinates when adding — not just the first entry.

Your working:

2. Let $x = (1, 2, 3)$ and $z = (1, 2)$. Explain why $x + z$ is not defined.

HINT Compare the number of coordinates in each tuple before attempting to add them.

CAUTION — before you solve Don't pad the shorter tuple or truncate the longer one to force the addition — the definition allows no such fallback.

Your working:

3. Let $x \in F^n$ and $c, d \in F$. Use a generic coordinate to prove $(c + d)x = cx + dx$.

HINT Fix an arbitrary index i and expand both sides using scalar multiplication in F^n , then field distributivity.

CAUTION — before you solve Remember to argue with a generic index i , not a specific n , and cite distributivity in F as its own step.

Your working:

Matrices — addition, scalar multiplication & equality

4. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -1 \\ 2 & 5 \end{bmatrix}$. Compute $A + B$ and $3A$. Then let $C = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$; explain why $A + C$ is not defined.

HINT Check that both matrices share the same shape before combining them entrywise.

CAUTION — before you solve Don't add a matching block of C to A and drop the extra column — mismatched shapes mean the whole sum is undefined.

Your working:

5. Let $c = 4$, $A = \begin{bmatrix} 2 & -4 \\ 6 & 0 \end{bmatrix}$, $B = \begin{bmatrix} \frac{1}{2} & -1 \\ \frac{3}{2} & 0 \end{bmatrix}$. Verify that $A = cB$ using the definition of matrix equality (i.e. check every entry of cB against the corresponding entry of A).

HINT Compute each entry of cB individually and compare it to the corresponding entry of A .

CAUTION — before you solve Don't just judge the matrices "proportional overall" by eyeballing — verify $A_{ij} = (cB)_{ij}$ entry by entry.

Your working:

6. Let $A, B \in M_{n \times k}(F)$ and $c \in F$. Prove $c(A + B) = cA + cB$ using generic entries A_{ij} , B_{ij} .

HINT Fix an arbitrary entry (i, j) and expand both sides using the definitions of matrix addition and scalar multiplication.

CAUTION — before you solve Remember a single generic entry (i, j) covers the whole matrix at once — no need to argue rows and columns separately.

Your working:

Matrix transpose

7. Let $A = \begin{bmatrix} 1 & 0 \\ 2 & -1 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 2 \end{bmatrix}$. Compute A^T , and determine whether $A^T + B$ is defined; if so, compute it.

HINT Transpose A first, note its new shape, then check that shape against B before adding.

CAUTION — before you solve Recheck compatibility using A^T 's new shape, not A 's original shape — transposing changes what it can be added to.

Your working:

8. Let $A, B \in M_{n \times k}(F)$. Prove $(A + B)^T = A^T + B^T$ using generic entries.

HINT Fix a generic entry (i, j) and apply the transpose definition to both sides of the equation separately.

CAUTION — before you solve Swap both the row and column index together when applying the transpose definition, not just one of them.

Your working:

Matrix-vector and matrix-matrix multiplication

9. Let $M = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$. Compute $M\vec{v}$ using the column picture, then verify your answer using the row picture.

HINT In the column picture, scale each column of M by the matching entry of $\vec{v}$ before adding them.

CAUTION — before you solve Don't add the raw columns unscaled, and be careful to match row entries with vector entries in the correct order.

Your working:

10. Let A be a 3×2 matrix, B a 2×4 matrix, and C a 1×2 matrix. Determine which of the products AB , BA , CB , BC , CA , AC are defined, and give the shape of each defined product.

HINT For each pair, check whether the number of columns of the first matrix matches the rows of the second.

CAUTION — before you solve Don't assume a product is defined just because the same number appears in both shapes — check columns-of-first vs. rows-of-second, in that order.

Your working:

11. Let $X = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $Y = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}$. Compute XY and YX . What does the result tell you about matrix multiplication in general?

HINT Use the row-dot-column rule for each entry rather than combining the matrices position by position.

CAUTION — before you solve Don't multiply entrywise (matching same positions) just because X and Y share a shape — use the row-dot-column rule instead.

Your working:

F -valued functions as vectors

12. Let $f(x) = x^2 + 1$ and $g(x) = 2x$. Compute $(f + g)(x)$ and $(3f)(x)$.

HINT Apply the definitions of function addition and scalar multiplication pointwise, term by term.

CAUTION — before you solve Remember to distribute the scalar to every term of $f(x)$, not just the leading term.

Your working:

13. Let $f(x) = x$ and $g(x) = |x|$. Are f and g equal as elements of $\mathcal{F}(F)$? Justify your answer using the definition of function equality.

HINT Test whether $f(x) = g(x)$ holds for every $x \in F$, including negative values.

CAUTION — before you solve Don't conclude equality just because the functions agree for $x \geq 0$ — pointwise equality requires agreement at every input.

Your working:

14. Let $f, g \in \mathcal{F}(F)$ and $c \in F$. Prove $c(f + g) = cf + cg$ using a generic input $x \in F$.

HINT Fix an arbitrary x and unfold the definitions in order: function addition, then scalar multiplication, then field distributivity.

CAUTION — before you solve Don't jump straight to the algebra shortcut — justify each equality (addition, scalar mult., distributivity) as its own step.

Your working:

More Practice, Graded by Level

15. (Level: Foundational) Let $A = \begin{bmatrix} 2 & 1 \\ -3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 2 \\ 0 & 5 \end{bmatrix}$. Compute $A - B$ and $2B$.

Your working:

16. (Level: Intermediate) Let $M = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$ and $N = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$. Compute MN and NM , and confirm $MN \neq NM$.

Your working:

17. (Level: Challenge) Let $f, g \in \mathcal{F}(F)$ with $f(x) = x^2$ and $g(x) = x + 1$. Compute $(f \circ g)(x)$ and $(g \circ f)(x)$, and determine whether $f \circ g = g \circ f$.

Your working:
